Lab Assignment 1

Title: In an e-commerce system, customer account IDs are stored in a list, and you are tasked with
writing a program that implements the following:
• Linear Search: Check if a particular customer account ID exists in the list.
• Binary Search: Implement Binary search to find if a customer account ID exists, improving the search efficiency over the basic linear

Program code:

```
#include<stdio.h>
void linear(int arr[], int n){
   printf("\n========Linear Search========");

   int key, i, flag=0;
   printf("\n\nEnter customer ID to search: ");
   scanf("%d",&key);

   for(i=0;i<n;i++)
   {
      if (arr[i]==key)
      {
         flag=1;
         break;
      }
   }
   if(flag==1)
   {
      printf("\nCustomer ID found at index %d",i);
   }
   else
   {
      printf("\nCustomer ID not found");
   }
}
void binary(int arr[], int n){
   printf("\n========Binary Search========");

   int key, i, flag=0;
   printf("\n\nEnter customer ID to search: ");
   scanf("%d",&key);

   int found=0;
```

```
    int low=0, high=n-1, mid;

    do{
        mid=(low+high)/2;
        if(key==arr[mid]) {
            printf("\nCustomer ID found at Index %d", mid);
            found=1;
        }
        else if(key>arr[mid]) {
            low = mid+1;
        }
        else {
            high = mid-1;
        }
    }while(found==0 && low <= high);

    if(found==0){
        printf("Customer ID not found");
    }
}
int main(){
    int arr[10], n, i;
    printf("\nEnter total no. of customers (max 10): ");
    scanf("%d",&n);

    printf("\nEnter customer IDs: ");
    for(i=0;i<n;i++)
    {
        printf("\nEnter ID of customer no. %d: ",i+1);
        scanf("%d",&arr[i]);
    }
    printf("\nEntered customer IDs are: ");
    for(i=0;i<n;i++)
    {
        printf("%d ",arr[i]);
    }

    printf("\n");

    char ch;
    do{
        int choice;
        printf("\nWhich type of search do you want to perform? \n1. Linear Search \n2. Binary
Search \nEnter choice (1 or 2): ");
```

```
        scanf("%d",&choice);

        switch(choice) {
            case 1:
                linear(arr, n);
                break;
            case 2:
                binary(arr, n);
                break;
            default:
                printf("\nEnter valid choice");
        }

        printf("\n\nDo you want to continue?(y/n): ");
        scanf("  %c", &ch);
    }while(ch == 'y'|| ch == 'Y');
    return 0;
}
```

Output:

```
Enter total no. of customers (max 10): 5

Enter customer IDs:
Enter ID of customer no. 1: 101

Enter ID of customer no. 2: 102

Enter ID of customer no. 3: 103

Enter ID of customer no. 4: 104

Enter ID of customer no. 5: 105

Entered customer IDs are: 101 102 103 104 105

Which type of search do you want to perform?
1. Linear Search
2. Binary Search
Enter choice (1 or 2): 1

========Linear Search========

Enter customer ID to search: 101
```

```
Customer ID found at index 0

Do you want to continue?(y/n): y

Which type of search do you want to perform?
1. Linear Search
2. Binary Search
Enter choice (1 or 2): 2

========Binary Search========

Enter customer ID to search: 104

Customer ID found at Index 3

Do you want to continue?(y/n): y

Which type of search do you want to perform?
1. Linear Search
2. Binary Search
Enter choice (1 or 2): 1

========Linear Search========

Enter customer ID to search: 106

Customer ID not found

Do you want to continue?(y/n): n
```